

WEEK 5



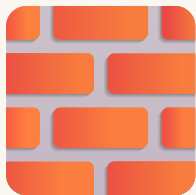
Roofed House

Cap the walls with a roof

벽 위에 지붕 올리기

BIG PICTURE

Walls + roof = a real house 🏠



Last week: walls. This week: a roof on top.

벽에 지붕을 더하면 집 완성

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Go up **first**, then build the roof

A roof must sit on **top** of the walls — so the agent climbs up before placing it.

```
move up by 1
repeat 3 times:
  place block down
  move forward
```

Without **move up**, the roof lands on the floor.

먼저 위로 올라가서 지붕을 놓아요

CONCEPT

A roof is a flat square layer

```
repeat 4 times:  
  repeat 4 times:  
    place block down  
    move forward  
  turn right
```

Nested loop again — but this time flat, like a lid.

지붕은 평평한 정사각형 층이에요

CONCEPT

Order matters



up → place

roof on the walls



place → up

roof on the floor

순서가 결과를 바꿉니다

CODE

Blocks: climb then roof

```
# stand on top of the walls first  
move up by 4  
repeat 4 times:  
    repeat 4 times:  
        place block down  
        move forward  
    turn right
```


CODE

How the teacher world builds it

```
# house roof – week 5 result  
blocks.fill(OAK_PLANKS, pos(10,4,0), pos(16,4,6))  
blocks.fill(OAK_LOG, pos(13,5,0), pos(13,5,6))
```

The **4** in the middle is the **height** — the roof sits at y=4, on top of the walls.

가운데 숫자가 높이(y)예요

PREDICT 🤖

Same height, or higher?

Predict first — then click reveal

```
move up by 1  
repeat 3 times:  
  place block down  
  move forward
```

Reveal output 🔍 reveal

VOCAB

New words today

roof the flat layer covering a house top

layer one flat level of blocks

on top of resting above something

order the sequence commands run in

지붕

층

위에

순서

Spot the difference

These should do the same thing — but one is buggy.

clean

```
repeat 4 times:  
  place block down  
  move forward  
turn right
```

buggy

```
repeat 4 times:  
  place block down  
  move forward  
turn right
```



DEBUG

The bug

Bug: in the buggy version `turn right` is **indented**, so it turns on *every* loop — the agent spins in a tiny circle instead of laying one straight side then turning once.

Indentation decides what's inside the loop.

YOUR TURN!

Build a roofed house

✓ Open this week's world (walls already built)

✓ Move the agent **up** to wall height first

✓ Use a nested loop to lay a flat roof

✓ Check no holes are left in the top

✓ Send a photo of your finished house



QUIZ

QUESTION 1 OF 3

Before placing a roof, the agent should:

Nothing — roofs build themselves.

Move up to the top of the walls.

Turn left four times.

Move down into the ground.



QUIZ

QUESTION 2 OF 3

Why does indenting `turn right` break the loop?

It makes the loop run forever.

Indenting never changes anything.

It turns on every repeat instead of once.

It deletes the blocks.



QUIZ

QUESTION 3 OF 3

Which order puts the roof **on the walls**?

move down, then place the roof.

place the roof, then move up.

move up, then place the roof.

Order doesn't matter.

QUIZ

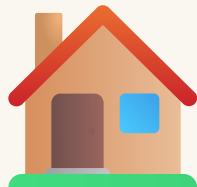
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



House complete!

Next: dig a well.

다음 주: 우물 파기